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ORIGINAL RESEARCH

VGA-Net: Vessel graph based attentional U-Net for retinal vessel segmentation

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Abstract

Segmentation is crucial in diagnosing retinal diseases by accurately identifiying retinal vessels. This paper addresses the complexity of segmenting retinal vessels, highlighting the need for precise analysis of blood vessel structures. Despite the progress made by convolutional neural networsks (CNNs) in image segmentation, their limitations in capturing the global structure of retinal vsessels and maintaining segmentation continuity present challenges. To tackle these issues, our proposed network integrates graph convolutional networks (GCNs) and attention mechansims. This allows the model to consider pixel relationships and learn vessel graphical structures, significantly improving segmentation accuracy. Additionally, the attentional feature fusion module, including pixel-wise and channel-wise attention mechansims within the U-Net architecture, refines the model's focus on relevant features. This paper emphasizes the importance of continuty preservation, ensuring an accurate representation of pixel-level information and structural details during sefmentation. Therefore, our method performs as an effective solution to overcome challenges in retinal vessel segmentation. The proposed method outperformed the state-of-the-art approaches on DRIVE (Digital Retinal Images for Vessel Extraction) and STARE (Structed Analysis of the Retina) datasets with accuracies of 0.12% and 0.14%, respecttively. Importantly, our proposed approach excelled in delineating slender and diminutive blood vessels, crucial for diagnosing vascular-related diseases. Implementation is accessible on https://github.com/CVLab-SHUT/VGA-Net.

1 | INTRODUCTION

The retina is characterized as a layer of light-responsive membrane that envelops the interior of the eyes. The primary responsibility of this layer is vision and includes components of the optic disc (OD), macula, and a network of blood vessels. Retinal images include rich tissues, such as retinal blood vessel structure, which can provide clinically significant information [1]. Retinal vessels are uniquely visible and can be observed directly and non-invasively, unlike any other component of the human circulatory system. They contain critical morphological characteristics such as diameter, length, and branching angle [2]. Therefore, by examining retinal vessels and extracting their features, it is possible to diagnose many diseases like diabetes, high blood pressure, arteriosclerosis, and cardiovascular diseases. The most important of these conditions are diabetes-related glaucoma and diabetic retinopathy (DR), which can result in vision loss leading to blindness, even if not diagnosed early [3]. Therefore, the timely detection of the disease significantly contributes to stopping or reducing the likelihood of its advancement.

The primary components within the eye are the retinal blood vessels, which are observable in colour fundus images. Taking these images has many benefits, such as helping to identify retinopathy, tracking the progression of diabetic retinopathy over time, and providing advice to patients. Colour fundus images serve as a vital tool for ophthalmologists in identifying and managing various eye diseases such as retinal micro-aneurysm, haemorrhage, and exudation [4]. Automated retrieval of retinal blood vessels from fundus images helps ophthalmologists a lot in the process of examining and diagnosing eye diseases. The manual examination is a lengthy and

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difficult process that requires high medical expertise. Although experienced ophthalmologists are responsible for manually segmenting retinal vessels, this approach is tedious and exhausting [5]. Automated retinal vessel extraction offers a potential solution to alleviate the workload on expert ophthalmologists, improve objectivity and consistency, and address the scarcity of experienced ophthalmologists in remote regions. Consequently, automated segmentation technology assumes a crucial role in facilitating the clinical diagnosis and treatment of diverse ocular conditions.

Numerous researchers have directed their attention toward automating the segmentation of retinal vessels. Nonetheless, researchers continue to encounter various challenges that make it a formidable task. The tree structure of the blood vessels within the retina is relatively complex because of the tiny and delicate blood vessels and their close connection with each other. Furthermore, the retinal target vessel in the image exhibits low contrast with the background, making accurate segmentation challenging. Also, various image foregrounds and backgrounds, contrasts, and colours have influenced retinal blood vessels. Moreover, the existence of noise in retinal images makes vessel segmentation significantly more difficult [6]. Consequently, accurate and high-quality segmentation methods are essential for analysing the initial condition and subsequent treatment of eye diseases.

The rise of deep convolutional neural networks (DCNN) has instilled optimism in tackling the above issues. According to [7], the segmentation process has observed remarkable progress with the utilization of convolutional neural networks (CNNs). However, these models are still in segmentation field due to poor continuity of small retinal vessels and insufficient segmentation. As a result, standard CNNs cannot optimally meet clinical diagnostic needs, and their performance requires improvement. Recently, numerous CNN models have been proposed to overcome these constraints. The fully convolutional neural network (FCN) [8] was among the pioneering deep neural networks employed for image segmentation. This model uses the deconvolution layer to restore the clarity of images. In this way, it solves the problem of semantic segmentation. Next, segmentation networks based on the encoder-decoder, such as SegNet [9], were proposed. These networks achieve pixel segmentation by down-sampling input feature maps to lower resolutions.

The U-Net model [10] stands out as the most renowned segmentation network within the category of encoder-decoder models. U-Net has unique encryption and decryption layers linked by skip connections, which integrate both the higher-level and lower-level characteristics of the image, leading to improved model accuracy and resolution of the vanishing gradient problem. Recently, numerous variations of the U-Net architecture have been suggested, such as U-Net++ [11], Attention U-Net [12], Dual Attention Network (DA-Net) [13], Res-UNet [14], and Bi-Directional ConvLSTM U-Net with Densely Connected Convolutions (BDCU-Net) [15].

Although U-Net family networks demonstrate strong segmentation performance, there exist several limitations: First, the convolutional operation only considers neighbouring regions, which may not effectively capture long-range semantic dependencies. Secondly, U-Net-based models struggle with accurately segmenting small and complex blood vessels in fundus images. Additionally, there exists a semantic gap between the decoder and encoder feature maps. Furthermore, these networks require high computational resources and may be less interpretable. Overfitting can occur when the dataset is small. Lastly, some U-Net variants may produce vessel maps with fractures.

In addition to the problems mentioned above, CNN-based methods are intrinsically capable of learning local features in images, but their ability to learn information about the global structure of images is limited. Furthermore, medical images often exhibit infinite spatial patterns integral to the diagnosis of diseases, yet methods relying on convolutional networks often encounter challenges in analysing these patterns. In addition, convolutional networks are often unable to maintain the continuity of the structure of the retinal vessels, while these vessels have a tree-like graphic structure that must be carefully analysed, and this issue is essential in the treatment of eye diseases. Existing CNN-based methods focus more on pixel-by-pixel performance and reduce the focus on assessing the graphical similarity of vessels. Learning the graphical structures of vessels not only enhances segmentation accuracy but also yields advantages in subsequent analysis steps [16].

To overcome these drawbacks, we introduced VGA-Net, a novel method based on graph convolutional networks (GCNs) and attentions. GCNs [17] have the capability to take into account the relationships between various pixels and learn the graphical structures of vessels. This can lead to improved accuracy in retinal vessel segmentation. In this way, our approach will have the ability to simultaneously extract pixel-level features and the relationships between nodes, resulting in more accurate segmentation. We use a CNN to extract features at the pixel level and a GCN to extract the relationships between different pixels and the graphical structures of vessels. Our proposed method uses the merits of CNN and GCN to jointly exploit pixel features and vascular graphic structures.

So far, many studies have not been investigated to learn the graphic characteristics of vessels. The problem with most GCN-based medical image segmentation methods that have been presented so far for vessel segmentation is that they have not addressed the issue of preserving the continuity of the vessel shape. Also, background pixels are usually ignored in generating graphs from images. Also, they have not been successful in segmenting the blood vessels in the central line of the retina and have suffered from over-segmentation. So, our goal in this paper is to overcome these challenges.

Our proposed technique initially breaks down the input fundus image into non-overlapping segments of a specified size, then considers each patch as a node in a graph. These nodes are connected with their 1-hop neighbouring nodes through vertical, horizontal and diagonal undirected edges. To extract the features of each node, a CNN based on architecture is used. Then, by applying a graph attention module to the node features matrix and the graph adjacency matrix, the graph features are extracted. Finally, the output of this module is fed into our designed segmentation U-Net-like network. Convolutional

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blocks in the network's decoder section extract rich feature representations at low levels, which are then merged with high-level graph information captured by the GCN modules. Then, the encoder produces the final predictions. In our segmentation network, we used a BConvLSTM module for integrating high- and low-level feature maps. Also, we have used pixel and channel attention modules in the decoder after each sampling layer above.

In this way, it is guaranteed that our proposed method preserves both the pixel information and the structural characteristics of vessels. Also, the continuity of the shape of the vessels is kept and finally, segmentation is done with high accuracy.

This paper makes significant contributions in the following aspects:

- Integration of GCNs: The incorporation of GCNs allows the model to consider relationships between pixels and learn the graphical structures of retinal blood vessels. This enhances accuracy in vessel segmentation by capturing long-range dependencies and addressing the complexity of vessel structures.
- Graph attention module (GAT): By applying attention mechanisms to node features and the adjacency matrix of the graph, the method can effectively extract relevant graph features. This attention mechanism contributes to a more nuanced understanding of the graphical representation, improving segmentation results.
- Continuity preservation: The explicit focus on preserving the continuity of vessel shapes is a key innovation. This addresses a common limitation in existing methods and ensures that the model maintains both pixel-level information and structural details, resulting in more accurate and clinically relevant segmentation.
- Designing an attentional feature fusion module: The incorporation of attention mechanisms, including pixel-wise and channel-wise attention modules within the U-Net, boosts the network's capability to emphasize relevant features, thereby enhancing both segmentation accuracy and the quality of feature representation.

The paper is structured in the following manner: In Section 2, an extensive overview of existing research in medical image segmentation is presented. In Section 3, the proposed approach is elaborated. In Section 4, we cover Experiment Data and Evaluation Metrics, discussing the datasets and metrics used for model performance evaluation. Section 5 focuses on Comparison and Ablation Study, where we compare our model with others and analyse the effects of each component. In Section 6, the overall conclusions are presented.

2 | LITERATURE REVIEW

This paper has organized vessel segmentation methods by their learning model, resulting in two primary types: unsupervised and supervised approaches.

2.1 | Unsupervised methods

An unsupervised method can identify and differentiate between the pixels representing vessels and non-vascular without prior knowledge [18]. This approach is applicable to perform data clustering. Unsupervised learning techniques are employed to explore unannotated data to detect patterns, which can group the data into subsets. These methods use artificial features, including edges, shapes, and spatial regions, to accurately extract vessels. Unsupervised segmentation techniques can be categorized into four subtypes: matched-filtered (MF) approaches [19, 20], mathematical morphology approaches [21, 22], model-based approaches [23, 24], and vessel tracking approaches [25, 26].

2.1.1 | Matched filter (MF) approaches

Chaudhuri et al. [19] initially proposed using MF for retinal vessel segmentation. They convolved retinal images with a two-dimensional Gaussian kernel to capture vessel features. However, their model's accuracy was not remarkable. Next, Sofka and Stewart [20] proposed a pixel-based segmentation algorithm, integrating multiple scales for precise vessel delineation. They combined essential features to train a classifier determining the "vesselness" of each pixel.

2.1.2 | Mathematical morphology approaches

Using mathematical morphology for extracting image structures, such as borders, skeletons, features, and convexity, constitutes geometric-based approaches. To accurately segment retinal vessels, several methods have been proposed. Miri and Mahlujifar [21] have proposed a technique that employs morphological processing of poly-structural elements and curve transformation. Their approach corrected the wrongly detected edges in the image, resulting in the advantage of well-segmented small blood vessels. Palomera–Pérez et al. [22] have presented a parallel implementation method based on a region-growing algorithm and multi-scale performance extraction. Their approach employed Gabor wavelet transformation and histogram equalization to strengthen the retinal vessels in the preprocessing stage. In the subsequent step, they applied growth level and grade methods separately, which yielded the outcome.

2.1.3 | Model-based approaches

Retinal vessel maps are often extracted using model-based approaches, which employ explicit vessel models. These methods encompass various techniques such as region growing, vector fields, active contouring and others. There are two general model-based methods: vessel profile models and deformable models. In one study, Espona et al. [23] employed deformable models and vessel topological properties to extract

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TABLE 1 Overview of unsupervised approaches.

Method	Pros	Cons	References
Matched filter (MF)	Simple, effective	Detects non-vascular edges, low accuracy	Chaudhuri et al. [19], Sofka and Stewart [20]
Mathematical morphology	Preserves image edges, removes noise	Time-consuming, requires human input	Miri and Mahlujifar [21], Palomera-Pérez et al. [22]
Model-based (vessel profile/deformable)	Simple implementation, low complexity	Needs human assistance for initial contours, less effective with concave edges	Espona et al. [23], Al-Diri et al. [24]
Vessel tracking	Extracts basic vessel info, efficient	Terminates at branching/crossing points, defects in results	Can et al. [25], Liu et al. [26]

retinal vessels. They used the outer curvature of the multi-level array to map the folds in the vessels and then initialized an active contour at the intersection of the vascular folds to trace near the optic nerve. The proposed algorithm exhibited an accuracy of 93.52%, as reported by the authors. Al-Diri et al. [24] also segmented retinal vessels using active contour. They first applied the tram line filter to segment some pixels of the central line of retinal vessels, then used the segment growth algorithm and dual active lines to initialize the vessel segments from these segmentation pixels.

2.1.4 | Vessel tracking approaches

The utilization of vessel tracking techniques is frequently employed to tackle issues related to examining vessels, for example, severe vessel stenosis, small size, low contrast, artefacts, and other related problems [25, 26]. For example, Can et al. [25] put forward a method that automatically identifies a reference point for blood vessel tracking as the minimum local grey level. It uses an iterative algorithm with direction patterns to find the maximum response. The bifurcation or intersection points are approximated, and the edges that respond the most to the patterns are highlighted in the candidate points. This approach is known for its robustness. Liu et al. [26] also proposed a vessel tracking algorithm for retinal vessel segmentation. Initially, the algorithm selects a search area based on vessel characteristics. Then, it identifies candidate edge points for the next vessel and classifies vessel structures into three categories. Using Bayesian theory, it computes the final edge points, demonstrating promising tracking results.

2.1.5 | A brief overview of unsupervised methods

Various methods for unsupervised vessel detection have distinct advantages and drawbacks. We have compiled an overview of the strengths and weaknesses of unsupervised methods in Table 1.

2.2 | Supervised methods

Compared to unsupervised techniques, supervised methods require the manual annotation of samples to construct a prediction model. There are two categories of supervised approaches: traditional supervised methods and deep learning approaches. Traditional supervised methods involve human experts designing features manually and extracting them through feature annotations, which requires significant financial and material resources as well as time and effort [27].

Compared to traditional supervised methods, the main goal of techniques based on deep learning is to train on vast amounts of data, enabling the deep neural network to extract more profound image characteristics and feature relationships [28]. As a result, research on algorithms based on deep learning is of significant importance, as these approaches are automatic and can minimize the expenses of manual segmentation. Deep learning algorithms have gained widespread use in recent years, becoming a standard method employed in several vision-related tasks, including recognizing text, identifying speech, and segmenting images semantically [29]. Researchers have recently introduced innovative and powerful architectures inspired by deep learning techniques. There has been a notable increase in the utilization of deep learning techniques in the segmentation field of medical images over the past few years. The FCN was the pioneering deep neural network developed for performing pixel-level image segmentation tasks. The FCN overcomes the limitations of CNN by employing the deconvolution layer to sample the last convolution layer's feature map. So, it resolved the issue of semantic segmentation for images.

2.2.1 | CNN-based methods

After the advent of FCN networks, encoding and decoding structures have been used to accomplish image segmentation tasks. SegNet [9] is one of the networks that can be mentioned in this category. These networks achieve pixel segmentation in the decoder section by sampling the input feature map with a lower resolution. However, despite their advantages, these networks suffer from certain limitations. One of the drawbacks is that extracting features is only combined at the final up-sampling stage, leading to inadequate utilization of multi-scale feature information during up-sampling. As a result, the model's capability to detect small details is reduced. A U-shaped network known as U-Net [10], inspired by the FCN network, has been introduced to overcome this issue. U-Net has been considered a significant advancement in segmenting medical images.

The U-Net network features a symmetrical structure, processing features from coarse to fine via contraction and expansion sub-networks. In the encoder stage, the input

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undergoes compression through convolutional layers and max-pooling, doubling feature channels at each down-sampling. In the decoder phase, up-sampling and convolutional layers rebuild the image, integrating features from adjacent layers. Skip connections address gradient disappearance, enhancing precision. The original U-Net has three main advantages: (1) it performs well with minimal training data, (2) it includes an encoding and decoding unit that can receive background information and locations simultaneously, and (3) it guarantees the whole texture of the input image.

Multiple methodologies utilizing U-Net-based approaches have been suggested for segmenting images in various domains. For instance, the U-Net++ [11] utilizes dense skip paths to incorporate scales with multiple levels within the skip connection. While the U-Net++ model is more proficient as comparing the conventional U-Net method, it has a high computational cost. Alom et al. [30] suggested the R2U-Net model, which merges U-Net with a recurrent residual convolutional neural network (RRCNN), to perform well in various datasets such as skin, lung, and blood vessels. However, when the dataset has a limited number of samples, R2U-Net suffers from overfitting. In another work, BDCU-Net, a bidirectional ConvLSTM U-Net network, was introduced for medical image segmentation [15]. BDCU-Net incorporates two innovations. The first is the use of dense convolutional layers within the final convolutional layer in the coding path of the U-Net architecture aids the network in obtaining a variety of feature maps rather than duplicative and redundant ones. The second innovation is using bidirectional ConvLSTM (BConvLSTM) in the skip connection path, which enables merging the feature map in a more complex way. BConvLSTM utilizes two ConvLSTMs to process data in both backward and forward directions, depending on the data's dependence. In another study, Wu et al. [31] suggested a network sensitive to scale and context for vessel segmentation. Although these algorithms are effective, they make the model more complicated and less interpretable. Also, more attention should still be given to thin vessels by these methods.

The attention mechanism has been newly introduced to enhance the capabilities of traditional deep learning models to perform tasks including segmentation, classification, and detection [32]. The primary concept of this mechanism involves allocating diverse weights to distinct features, allowing it to emphasize relevant features. The use and integration of the attention mechanism with deep neural networks can be very effective. Attentional mechanisms have been integrated into semantic image segmentation networks with positive results. For instance, the Attention U-Net [12] was introduced to exploit salient features in medical images. The network employs an integrated attention gate to focus on relevant regions and obtain richer contextual information automatically. However, U-Net attention suffers from a lack of deep monitoring and focus when there is a significant semantic gap between the two paths of network contraction and expansion, leading to an inability to record the characteristics of different levels in the decoder path.

The attention mechanism has gained popularity as a technique in the past few years among researchers. An example of this can be seen in the work of Xiao et al. [14], who put forward a model for segmenting vessels by integrating the attention mechanism in skip connections. Another model, squeeze-and-excitation networks (SE-Net) [33], was introduced to automatically determine the significance of individual feature channels by leveraging squeeze-and-excitation, which enhances useful information and suppresses useless information. The channel-attention U-Net [34] implemented a channel attention module (CAM) to extract critical semantic features from each level by leveraging higher-level features. However, this model did not incorporate spatial attention despite containing lower-level characteristics generated from the uppermost to the lowermost part. On the other hand, Spatial Attention U-Net (SA-UNet) [35] was introduced to address this issue. It focuses on the more informative pixels of the image using a spatial attention module, which suppresses unimportant pixels like background pixels. Unfortunately, SA-UNet has no channel attention and struggles at the point where narrow and wide blood vessels meet, resulting in poor performance.

The authors of the Dual attention network (DA-Net) proposed a network that learns channel and spatial feature correlations [13]. However, this approach increases model complexity. An efficient channel attention network (ECA-Net) was introduced to address this issue. It uses an ECA module to exchange cross-channel information without dimensionality reduction, reducing model complexity while maintaining performance [36]. The IBA-U-Net, the attentive bidirectional convolutional LSTM with redesigned inception, was presented as an alternative segmentation approach [37]. The authors substituted the 3×3 convolutional layers in U-Net with the redesigned inception block. They used a combination of attention and BConvLSTM in the skip connection to highlight salient features. The segmentation outcomes achieved through the IBA-U-NET method were satisfactory for retinal vessels, skin lesions, and lungs. In another study, Kar et al. [38] explored retinal vasculature extraction using MSR-Net, a multi-scale residual CNN, in conjunction with a generative adversarial network (GAN) [39]. Preprocessing involved applying the contrast limited adaptive histogram equalization (CLAHE) method [40] to enhance blood vessel contrast. In the GAN framework, both a generator for segmentation and a discriminator for classification were integrated. Utilizing an inception module, the objective of the study was to capture detailed vessel segments by extracting multi-scale features at different scales. The discriminator was comprised of two layers of self-attention networks stacked on top of each other, along with a fully connected feed-forward network layer that processed positional information, resulting in a binary output. By harnessing the attention mechanism of the transformer, the model efficiently discriminated and preserved both local and global information.

Although there have been advances in segmentation techniques, they are encountering challenges. One of the challenges is that retinal fundus images are inevitably affected by diverse factors, such as uneven lighting during imaging, which can impair the model's performance when segmenting small blood vessels, especially at intersections of thin and thick blood vessels. Additionally, the essential constraint of techniques utilizing CNNs are only capable of modelling local information. So, it

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is challenging for them to consider global structural information. To overcome this problem, researchers have introduced graph neural networks (GNNs) [41] as a solution to perform convolutional operations tailored for graphs. GCNs [17], a subset of GNNs, have been recognized as effective methods for analysing data with known graph structures. GCNs leverage CNNs to encode both the local connections within graphs and the intricate characteristics of individual nodes within structured data. In contrast to typical convolutional neural networks, which depend on Euclidean distances for their operations, extracting features from data with non-Euclidean structures is a strength of GCNs.

2.2.2 | GCN-based methods

Numerous researchers have recently utilized the GCN architecture in medical image processing to address the inherent limitations of convolutional networks. For instance, Lu et al. [42] introduced a cutting-edge image segmentation approach known as Graph-FCN, incorporating a graph structure inspired by FCNs. The approach yielded superior results and marked the first application of GCNs to image segmentation. Shin et al. [43] introduced graph complexity through the creation of a graph representation for retinal vessel classification, where nodes were defined based on the vessel sample's skeleton. The method involved extracting edge information from the vessel skeleton and sampling node features from CNN feature maps. Yu et al. [44] maintained the topology of vessels by creating edges and predicting links between various nodes, which were generated from semantic information extracted using U-Net. Hao et al. [45] presented a new framework called EAGC_UNet++, a graph-based edge attention gate, for medical image segmentation. By leveraging a combination of a residual graph convolution network, UNet++ and edge attention gates (EAG), the proposed model has demonstrated significant improvements in the accuracy of medical image segmentation and tackled the problem of missing edges.

Despite the effectiveness of the GCN-based methods mentioned above, there are still some problems with them. First, the majority of these approaches prioritize pixel-level performance and allocate comparatively less consideration to assessing the graphical resemblance of vessels. The existing methods have not fully utilized the potential of GCNs, because they have failed to consider background pixels in image processing and graph creation. This can result in the loss of importatnt information. Also, these approaches have failed to preserve the structural continuity of vessels during segmentation.

To address the issue of maintaining vessel continuity, Yeganeh et al. [46] introduced a structural continuity preserevation method for medical image segmentation called SCOPE. This network utilizes graphs to maintain continuity and anatomical connectivity in medical image segmentation. This also ensures the preservation of predicted continuity by encoding the shape's continuity as a graph constraint. Although their proposed method is effective, a notable drawback is the increased size of connections, leading to over-segmentation. This adversely impacts the central line connection, resulting in a failure in segmentation. We have collected all the studies conducted in this section, along with their strengths and weaknesses, in Table 2.

To solve the mentioned limitations and inspired by the above studies, we have presented a method of combining CNN and GCN. Our method first segments the images and uses a CNN to extract the pixel level information of images. Then the CNN output is fed to an attention-based GCN module to extract the structural information and inter-pixel relationships of the vessels as well. Finally, the output of this module is fed to the skip connections in the U-Net-based segmentation network so that the pixel features are combined with the vascular structure and the segmentation work is done with higher accuracy. The details of the proposed approach are outlined in Section 3.

3 | PROPOSED METHOD

The specifics of our suggested methodology, dubbed VGA-Net, are outlined in this section. Initially, we provide a summary of the proposed framework. Subsequently, we delve into the specifics of each step and block presented in the framework.

3.1 | Framework overview

Our suggested approach comprises three phases: preprocessing, information extraction, and vessel segmentation by VGA-Net. Figure 1 shows our proposed general pipeline for retinal vessel segmentation. Also, the process is outlined as follows.

The blood vessels within the retina appear to have dark colour and elongated, and thin structures with different thicknesses. Alternatively, narrow vessels and the retinal background have a similar colour close. For this reason, aiming to enhance the contrast of blood vessels and the visual coherence of fundus images, it is essential to carry out preprocessing as the initial step [47]. In the next step, information needs to be extracted from these images at both the pixel and the structural level. Therefore, to extract information at the pixel level, the processed images are first divided into smaller patches, and each patch is considered as a node in a graph. Then, a CNN is used to extract pixel-level features. Subsequently, for extracting structural and graph-level information from the images, a graph convolution module (GCM) is employed. The features extracted from this module enable the model to extract relevant information from each image patch and the relationships between patches. Finally, the output of this stage is fed into the segmentation network. This helps the model to better identify patterns in new images and improves the generalizability and performance of the model in new cases.

3.2 | Data preprocessing

Preprocessing is the first and essential step for automatic retinal disease detection. One limitation of retinal imaging is the

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TABLE 2 Overview of previous approaches.

Traditional supervised methods	Accuracy	Requires manual annotation, expensive and time-consuming	Kumar et al. [27]
CNN-based methods			
FCN, SegNet, U-Net	Automatic, minimal training data, background info and location, preserves texture	Limited feature extraction at up-sampling, sensitive to uneven lighting	Long et al. [9], Ronneberger et al. [10]
U-Net variations (U-Net++, R2U-Net, BDCU-Net)	Improved performance over U-Net, dense skip paths, recurrent connections	High computational cost, overfitting with limited data, complex model	Zhou et al. [11], Alom et al. [30], Azad et al. [15]
Attention U-Net	Focuses on relevant features	Lack of deep monitoring, poor performance at semantic gaps	Oktay et al. [12]
Attention mechanisms (SE-Net, Channel/spatial attention U-Net)	Selects important features, suppresses useless info	Spatial attention missing in Channel U-Net, poor performance at vessel junctions in Spatial U-Net	Hu et al. [33], Yu et al. [34], Li et al. [35]
Dual attention network (DA-Net, ECA-Net)	Learns channel and spatial correlations, efficient	Increased model complexity	Fu et al. [13], Liu et al. [36]
IBA-U-Net	Redesigned Inception block, Selects important features and highlight salient features	The complexity of the model and low flexibility in dealing with different data	Yu et al. [37]
MSR-Net	Detecting fine vessel segments across multiple scales Capability to enhance accuracy in identifying retinal vessels	Higher computational resources and longer training time	Kar et al. [38]
GCN-based Methods			
Graph-FCN	First application of GCNs to image segmentation	Not widely adopted	Lu et al. [42]
GCN-based methods	Improved accuracy, addresses missing edges	Focus on pixel performance, ignores background pixels, doesn't maintain vessel continuity	Shin et al. [43], Yu et al. [44], Hao et al. [45]
SCOPE	Maintains vessel continuity	Over-segmentation, impacts central line connection	Yeganeh et al. [46]

Abbreviations: BDCU-Net, Bi-Directional ConvLSTM U-Net with Densely Connected Convolutions; FCN, Fully Convolutional Network; R2U-Net, Recurrent Residual Convolutional Neural Network; SE-Net, Squeeze-and_Excitation Networks.

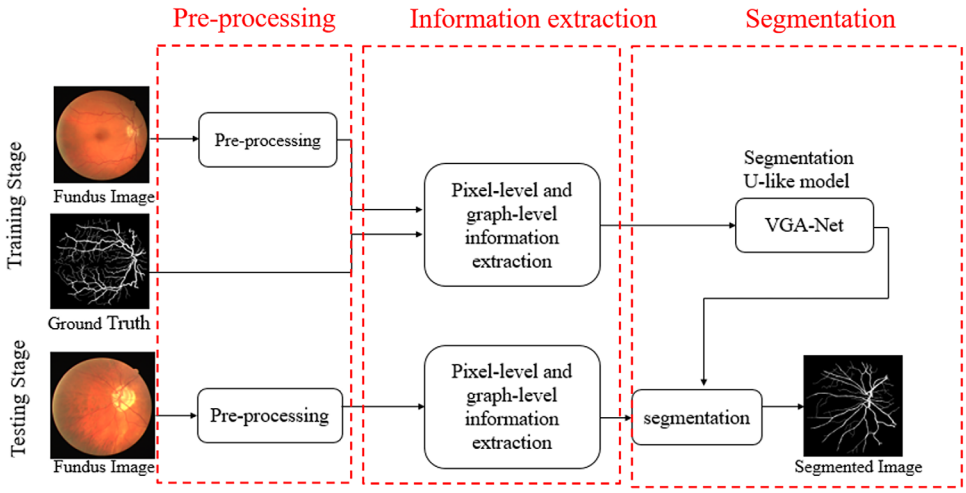


FIGURE 1 Overall structure of pipeline of proposed framework. VGA-Net, Vessel graph based attentional U-Net for retinal vessel segmentation.

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FIGURE 2 Preprocessing steps.

frequent suboptimal quality of the resultant images. To achieve segmentation with high accuracy, our goal is twofold: firstly, to enhance the quality of the fundus images, and secondly, to distinguish the vessels from the image background. Our objective is to improve the sharpness and detail of the fundus images through preprocessing techniques. The preprocessing consists of some steps described as follows:

(A) Background intensity normalization: Background intensity is a major challenge in retinal vessel segmentation. To address this challenge, we propose a method to normalize the background intensity by replacing the black areas with the average colour of the retina. This approach enhances segmentation accuracy by facilitating better differentiation between the background and the blood vessels. The step is optional.

(B) CLAHE: As mentioned before, distinguishing the boundary between the background and the vessels within fundus images poses a significant challenge. In order to improve the quality of the images, two commonly used methods are employed: the luminance enhancement method and CLAHE [40]. While global histogram equalization enhances noises placed on a homogeneous background, CLAHE enhances the contrast block. This paper utilizes the CLAHE technique to augment the contrast of fundus images. This approach improved the separation between the vessels and the background and highlighted the areas that contain the vessels in the normalized fundus image.

(C) Un-sharp masking filter: The un-sharp masking filter (UM) is a method for highlighting the prominent features of images, mainly their textures. First, a blurry image is used to generate a mask based on the original image. Subsequently, this mask is merged alongside the original image, subtracting the blurred version of the image [48]. By applying this filter, the edges in the image are emphasized, and an impression of a highly detailed image is produced. Figure 2 demonstrates the enhancement of fundus images using the un-sharp masking filter after CLAHE.

Finally, the background is converted back to black to restore the original appearance of the image. Figure 2 depicts the preprocessing steps as mentioned.

3.3 | Pixel-level and graph-level information extraction

Despite the advantages of CNNs in feature extraction, they have two main drawbacks. First, they are unable to consider the relationships between different pixels. This can lead to incorrect vessel segmentation, especially in areas where vessels are close together. Second, they are unable to consider the graphical structures of vessels. Retinal vessels often have a tree-like structure. This structure can be used to identify and segment individual vessels. To address the mentioned limitations, we propose a method based on the integration of CNNS and graph GCNs. GCNs can consider the relationships between different pixels and learn the graphical structures of vessels. This can lead to improved accuracy in retinal vessel segmentation. In this way, our approach will have the ability to simultaneously extract pixel-level features and the relationships between nodes, resulting in more accurate segmentation. We use a CNN to extract features at the pixel level and a GCN to extract the relationships between different pixels and the graphical structures of vessels.

In our proposed method, initially, the input image undergoes partitioning into non-overlapping patches of size $\mu \times \mu$, with each patch being considered a node within a graph. Next, a CNN based on the DRIU architecture is employed to extract pixel-level features. Subsequently, the node features are extracted using a graph attention module (GAT), capturing patterns and relationships among neighbouring nodes with varying weights. Finally, the output of this module is passed into the VGA-Net model for segmentation. Overall, this approach ensures the preservation of both pixel-level and structural information of vessels, leading to accurate segmentation results. Our workflow is as follows:

1) Graph construction (G): Initially, the input image undergoes a process where it is divided into non-overlapping patches of size $\mu \times \mu$. Each of these patches is assigned a node within a graph representation. These nodes are interconnected via undirected and unweighted edges, forming a complete graph structure characterized by an adjacency matrix A.
2) Pixel-level feature extraction: The pixel features of the image are used to generate the features of nodes (f_i). Therefore,

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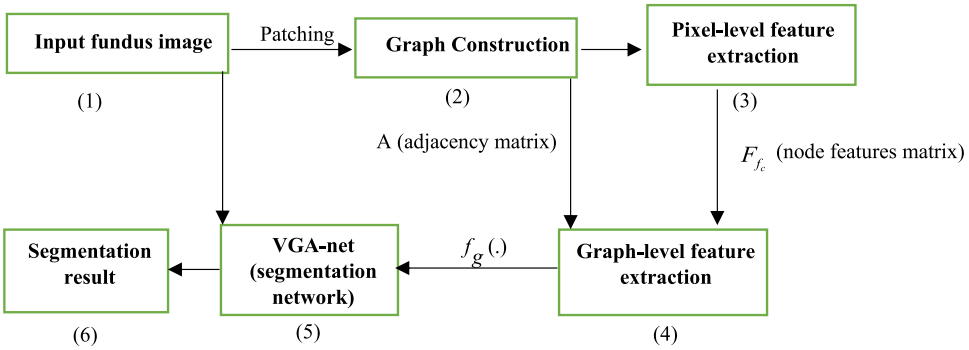


FIGURE 3 Overview block diagram of proposed method.

we use a CNN inspired by the VGG16 architecture. We adjust this network by replacing the convolution layer with a max-pool layer of dimensions $\mu \times \mu$ and pitch μ, where μ represents the patch size. We denote the matrix of features obtained from this stage as F_{f_c}.

3) Graph-level feature extraction: During this phase, the matrix of node features and the graph's adjacency matrix are passed to the graph convolution module (f_g). This module extracts graph-level node features that encompass relationships and patterns among neighbouring nodes. The GAT is employed for this purpose, allowing nodes to learn important information from their neighbours with varying weights through the attention mechanism. This feature enhancement improves model performance, particularly when the precise geometric structure of the graph is ambiguous or less significant.
4) Segmentation: During the segmentation stage, we reshape the output from the GAT and inject it into each skip connection of the decoder section of our U-Net-like network. This process enables the fusion of graph information with the feature map of each decoder stage. Subsequently, the combined information is passed to the encoder for further processing. This integration ensures that both graph-based and pixel-level features are effectively utilized throughout the segmentation process, leading to enhanced segmentation accuracy and the preservation of structural information.

Figure 3 offers an overview of the workflow, while Figure 4 elaborates on the stages of our proposed method in detail.

To gain a deeper understanding of the proposed method, we will now offer a comprehensive explanation of each stage illustrated in Figure 4.

(a) Graph construction: The input image $x \in R^{H \times W}$ is divided into a grid of non-overlapping patches of size $\mu \times \mu$. Every patch is regarded as a vertex within the graph. Each node is linked to its adjacent vertices within a one-hop distance on the same level by undirected and unweighted edges. The connections include vertical, horizontal, and diagonal connections. A direct edge construction strategy is used, in which each node is connected to all of its 1-hop neighbouring nodes around its patches.

(b) Pixel-level feature extraction: For extracting pixel-level features from images, we utilize a CNN based on DRIU architecture. DRIU [49] is built upon the VGG-16 network [50]. We receive a multi-level feature representation from the VGG stack, capturing both local and global pixel-wise features. To generate node-level features for our graph, we adapt the DRIU by replacing the 1×1 convolution with a max-pool layer of $\mu \times$ ¯ window and stride of ¯ where μ is the patch size. By applying this operator, we obtain a more compact and efficient feature representation for each node, which is crucial for training our segmentation model. This step produces a node feature matrix known as F_{f_c}, which exhibits dimensions of $N \times 64$, where N represents the number of nodes and 64 signifies the dimensionality of features. This matrix encapsulates the initial features of nodes extracted from the input image.

(c) Graph-level feature extraction: In this step, the output node feature matrix F_{f_c} from the previous step and the graph adjacency matrix A are passed to the GCM. GCM extracts graph features that capture the relationships and patterns between neighbouring nodes. One of the challenges of GCN is its strong reliance on the geometric structure of the graph. This means that we need to know how nodes are ordered or where they are positioned. In contrast, GAT [51] addresses this issue by utilizing attention to neighbours. It allows each node to accurately learn important information from its neighbours with different weights, without being dependent on a specific order or position. This feature leads to improved model performance, especially when the exact geometric structure of the graph is unclear or less significant. This paper utilizes, GAT to extract graph-level node features from a fully-connected graph $G = (V, E)$ with $V = \{v_1, \ldots, v_n\}$ representing node features $v_i \in R^D$ and edge set E. Attention coefficients are calculated and normalized using the Softmax function as follows:

$$e_{ij} = a(W_q v_i, W_k v_j) = \frac{W_q v_i (W_q v_j)^T}{\sqrt{D}} \quad (1)$$

$$a_{ij} = Softmax(e_{ij}) \quad (2)$$

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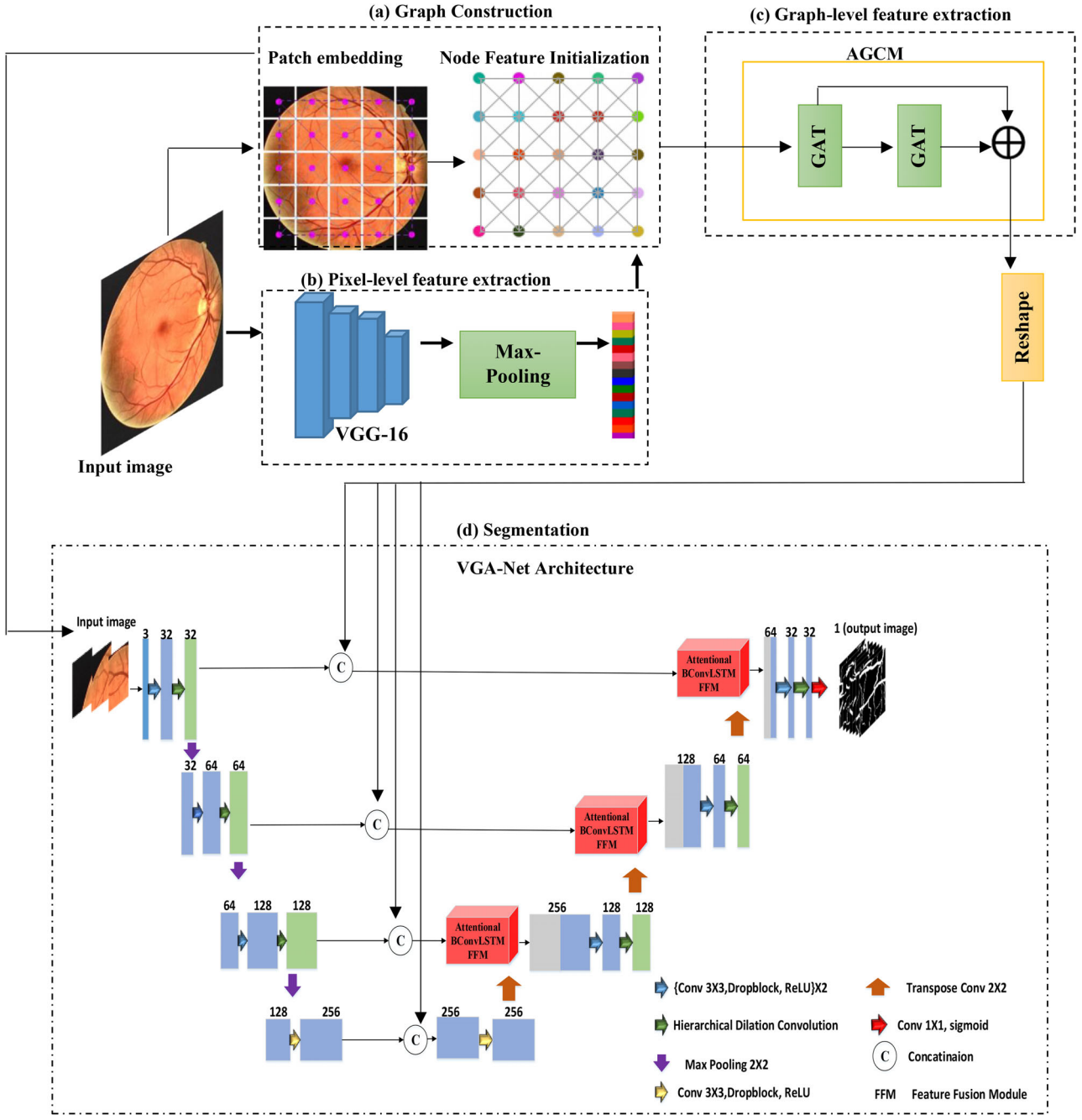


FIGURE 4 Detailed steps of the proposed method. AGCM, Attentional graph convolutional module; GAT, Graph attentional module; FFM, Feature fusion module.

Here, W_q and W_k are learnable parameters. To reduce computational expenses, in line with prior research [52], we substitute the single-layer feed-forward neural network with multi-head (MH) dot production, a more expedient and space-efficient alternative in practical applications.

$$MH(v_i, v_j) = \|_{h=1}^{H} (head_1, \ldots, head_h) W_o \tag{3}$$

$$head_h(v_i, v_j) = Att\left(W_q^h v_i, W_k^h v_j, W_v^h v_j\right) \tag{4}$$

$$Att(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{D}}\right) V \tag{5}$$

In the above equations, || denotes a concatenation operation and W_v^h, W_k^h, W_q^h and W_o are all with dimensions $D \times D$. Specifically, we utilize eight parallel attention layers ($H = 8$), resulting in d being one-eighth of D. Subsequently, employing a nonlinear activation function, the final output feature is calculated. Subsequently, utilizing a ReLU non-linear activation on the MH output, we incorporate batch normalization (BN)

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within the graph attention module to expedite the training process, allowing for the computation of the final output feature. At this point, the construction of the graph attention module is complete.

$$v_i' = BN\left(\mathrm{Re}\,LU\left(\sum_{j\in N_i}^{n} MH(v_i', v_j)\right)\right) \quad (6)$$

(d) Segmentation: Finally, the output of the graph attention module undergoes reshaping and will be fed into the VGA-Net model.

3.4 | VGA-Net model

The limitations of U-Net and its subsequent extensions can be summarized in several points: 1) There is a lack of long-range dependencies for convolution operations; 2) as the network depth increases, there is a significant degradation in the level of detailed information in the image; 3) there is inadequacy to extract features of long dependencies; and 4) there is a loss of global context features in the segmentation of complex medical images. Therefore, to overcome these issues, we designed a new U-Net structure called VGA-Net. Figure 4d shows this architecture. The proposed model primarily comprises HDC module, attentive BConvLSTM module. The attentive BConvLSTM module contains the spatial attention module referred to as SAM, the channel attention module known as CAM, and the bidirectional ConvLSTM.

3.4.1 | Dropblock method

As explained earlier, most existing fundus datasets contain an inadequate number of images. In other words, despite the increase in data before training the network, there is a concern regarding the potential occurrence of overfitting during the model's training. For this reason, it is needed to somehow prevent the existence of more overfitting. The dropblock technique can be considered a structured form of the dropout method. It can avoid overfitting problems in the network. Unlike dropout, dropblock drops a small patch of pixels randomly within the feature map. The dropblock module can completely solve network redundancy. Also, the effectiveness of the batch normalization (BN) technique and ReLU activation function effectively has been well established in reducing the duration for the network to converge the fundamental convolution unit employing dropblock [53].

3.4.2 | Hierarchical dilated convolution module (HDC)

The proposed network should be able to consider the many small, tiny, and fragile vessels that are present in the fundus images. Because these vessels are important in the identification of retinal disorders, segmenting vessels is a difficult task, especially when the vessels are very small. In this section, we introduce a module to help in the diagnosis and adequate segmentation of retinal vessels. The idea is to use dilated convolutions [54]. This idea can be used to compensate for the reduced feature map while using max-pooling in the encoding phase of the U-Net architecture. In the concept of using a hierarchical structure, first, the initial feature map, titled F, is divided into a pair of distinct feature maps named A1 and A2 in parallel along the channel axis. Then, convolution feature transformation is applied to these two feature maps in two spaces and at different scales. In this way, the possibility of recognizing information on different scales of blood vessels is realized. With the help of dilated convolutions, it is possible to identify the information on different scales of vessels. They are employed to obtain details regarding the configuration of vessel edges. Then, the feature maps B1 and B2 were obtained with corresponding dilation rates of 1 and 2. Finally, these two transformed feature maps are combined along the dimension of the channel, and the final new feature map called B3 is formed. Figure 5 depicts the schematic representation of the proposed approach.

3.4.3 | Attentional BConvLSTM feature fusion module (AB-FFM)

In this section, we've developed a feature integration block to improve the merging of low-level semantic details (X_d) with high-level semantic features (X_e) for reconstructing predictive diagrams. X_e and X_d are input sequentially. Our proposed approach utilizes BConvLSTM to encode these two feature maps, employing two ConvLSTM units to handle input data bi-directionally, accounting for both backward and forward directions. In this way, the gap in contextual information between the feature maps of the encoder and decoder is significantly reduced [55].

As depicted in Figure 6, X_e and X_d are combined through the BConvLSTM module. To prevent any adverse impact on prediction processes due to the varying semantic discrepancies, it is necessary to better capture global and local information from two feature maps. This can increase the network's sensitivity to important features, enhance network representation power, and improve network performance. One potential solution involves the development of distinct modules, one for channel attention and the other for pixel attention. So, merged features then enter a dual-attention network to autonomously discern the significance of integrated features at both the pixel and channel levels. The pixel-wise attention mechanism draws inspiration from channel attention [34] and spatial attention [35], initially introduced in the AIM 2020 challenge [56].

In the pixel-wise attention module (PAM), feature map F undergoes a 3×3 convolutional layer and max pooling to create an attention map. The attention map is subsequently transformed using the sigmoid function to derive pixel-wise weight scores. In the channel-wise attention sub-module (CAM), feature map F undergoes a 3×3 convolutional layer, succeeded by a global average pooling (GAP) layer. After this step, a reshaping operation is applied, followed by two

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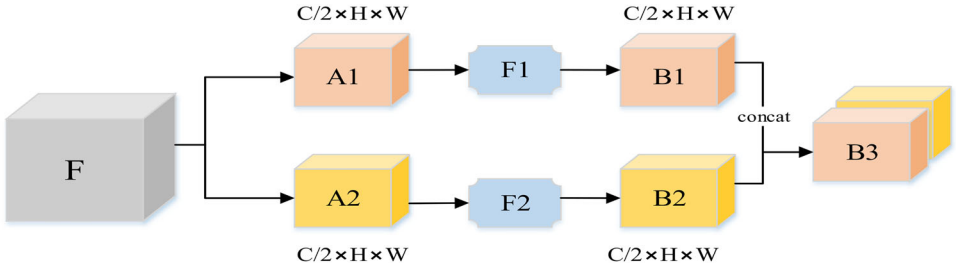


FIGURE 5 The overall structure of the hierarchical dilated convolution module.

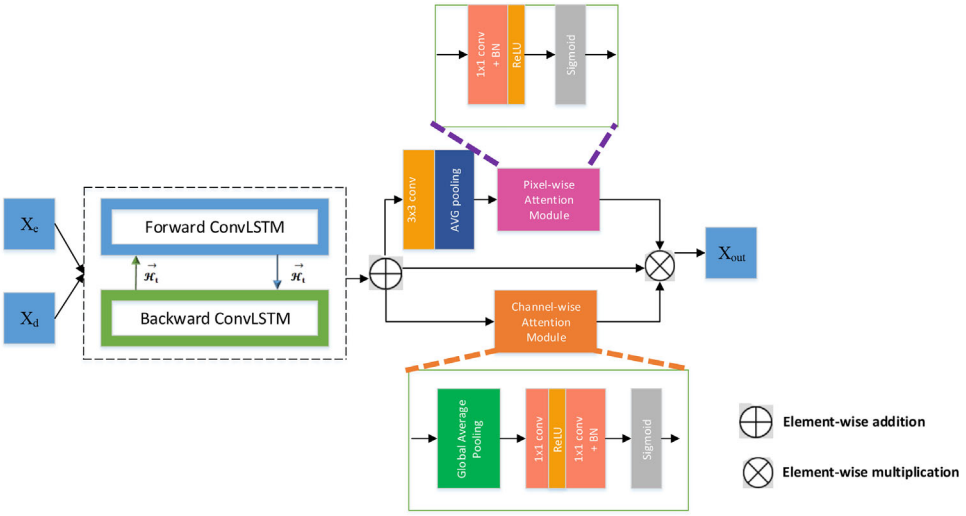


FIGURE 6 The structure of AB-FFM. BN, Batch normalizatio.

additional convolutional layers with ReLU and Sigmoid activation functions, an up-sampling operation, and finally, a multiplication operation to produce channel feature weight scores. In a process called AB-FFM in Figure 6, pixel-specific and channel-specific weight maps are applied to a feature map F, resulting in a new output that effectively reduces noise while amplifying vascular information.

4 | EXPERIMENT DATA AND EVALUATION METRICS

In this section, we elaborate on the experimental setup, datasets utilized, and evaluation metrics employed to evaluate the effectiveness of our proposed approach.

4.1 | Experiment data

For our study, we employed two publicly-accessible datasets comprising retinal fundus images of blood vessels, namely DRIVE[1] and STARE[2]. The DRIVE dataset, which stands for digital retinal images for vascular extraction, comprises 40 retinal images with accurate blood vessel segmentations. The Ophthalmic Image Analysis Group at Utrecht University, in the Netherlands, developed this dataset. The training set consists of 20 designated images, and an additional set of 20 images is set aside specifically for the test. The images have 565 pixels in width and 584 pixels in height.

The STARE dataset, which stands for structural analysis of the retina, is a freely accessible resource for researching retinal image analysis. This dataset comprises 20 digital fundus images. Each image is sized at 700 pixels in width and 605 pixels in height in terms of resolution. Additionally, the dataset has manual segmentation of vessels and optic discs for each image. These annotations were created through manual labour by two experts. Table 3 summarizes the required information from these two datasets.

TABLE 3 Summary of dataset information.

Dataset	DRIVE	STARE
Image number	40	20
Image size	565 × 584	700 × 605
Patch size	48 × 48	48 × 48
Patch number	190,000	181,500

[1] https://drive.grand-challenge.org
[2] http://cecas.clemson.edu/~ahoover/stare/probing/index.html

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4.2 | Evaluation metrics

This section employs various evaluation measures to assess the proposed model from different angles quantitatively. The measures used include accuracy (ACC), which represents the percentage of accurately predicted pixels; specificity (SP), which reflects correct background detection; sensitivity (SE), which indicates the proper identification of blood vessels and dice score (Dice). These metrics offer a thorough assessment of the performance of our approach, reflecting pixel-wise accuracy from multiple perspectives.

Due to a notable imbalance, metrics at the pixel level are insufficient in revealing clear distinctions. In evaluating the graphical coherence of vessels, Shit et al. [57] introduced the centreline-Dice (clDice) loss, aiming to motivate segmentation models to preserve tubular structures. This metric is used to evaluate the accuracy and sensitivity of a topological algorithm. It calculates both sensitivity and precision simultaneously. The topology precision metric quantifies the ratio of extracted skeletons by the algorithm to the skeletons present in the ground truth mask. Topology sensitivity, on the other hand, represents the ratio of skeletons in the ground truth mask to those extracted by the algorithm. In this study, we also utilize clDice as a metric to assess the connectivity and topology of vessel segmentations. The formula for clDice is as follows:

$$\text{clDice} = \frac{2 \times \text{Topology Precision} \times \text{Topology Sensitivity}}{\text{Topology Precision} + \text{Topology Sensitivity}} \tag{7}$$

where topology precision (TP) and topology sensitivity (TS) are defined as follows:

$$\text{Topology Precision (TP)} = \frac{\text{Number of extracted skeletons by algorithm}}{\text{Number of skeletons in ground truth mask}} \tag{8}$$

$$\text{Topology Sensitivity (TS)} = \frac{\text{Number of skeletons in ground truth mask}}{\text{Number of skeletons extracted by algorithm}} \tag{9}$$

Given the non-uniform distribution of blood vessel and non-blood vessel pixels, traditional evaluation metrics like accuracy (ACC), Dice coefficient (Dice), and class-balanced Dice (clDice) may not adequately capture the performance of the model. In such cases, the Matthews correlation coefficient (MCC) serves as a superior measure, particularly for imbalanced classes. MCC [58] considers both true positives, true negatives, false positives, and false negatives, providing a balanced assessment of classification performance, making it well-suited for scenarios where class distributions are skewed. Therefore, MCC is preferred for evaluating models in situations where imbalanced classes are present, such as in the segmentation of blood vessels in medical images. The Matthews correlation coefficient (MCC) formula is defined as:

$$\text{MCC} = \frac{(\text{TP} \times \text{TN} - \text{FP} \times \text{FN})}{(\text{TP} + \text{FP})\,(\text{TP} + \text{FN})\,(\text{TN} + \text{FP})\,(\text{TN} + \text{FN})} \tag{10}$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives.

4.3 | Details of implementation

We conducted the experiments using PyTorch 1.13.1 on a Windows operating system. We utilized a GTX 1080Ti GPU for our experiments. The training process involved utilizing patches obtained from the fundus images of two distinct databases, with a training-validation-testing split of 70%, 20%, and 10%, respectively. During the training phase, our experimentation on the DRIVE dataset involved employing a batch size of two and training the course of 100 iterations, while utilizing a learning rate setting to 1×10^{-3}. When working with the STARE dataset, we adopted these values 3, 80, and 1×10^{-1}, respectively. All of these values precisely match those specified in [34]. Additionally, papers referenced as [59] and [60] also confirm some of these parameter values.

We utilized the Adam optimizer during training, setting the learning rate to 0.0001, and if the validation loss remained unchanged for five successive epochs, early stopping was implemented. The network was trained using dice loss as the loss function. Additionally, for the DropBlock, we established a discarded block size of 7 and implemented a dropout rate of 0.15. During the testing phase, the model parameter value was 973,125, and the parameter size was approximately 3.97 M.

5 | COMPARISON AND ABLATION STUDY

Within this section, before proceeding with the evaluation of model components, we explore the impact of image contrast enhancement methods on segmentation performance through ablation experiments. Then, we performed ablation experiments to evaluate the efficacy of the various components employed in our proposed model. These experiments were performed at two levels: the first level aimed to assess the effectiveness of vessel segmentation at both pixel and graph levels. Following these experiments at both levels, we comprehensively compared and analysed our proposed method with

5.1 | Ablation study on image contrast enhancement methods

Before delving into the analysis of the model components, we conducted an ablation study on image contrast enhancement methods to evaluate their influence on segmentation performance. In the pre-processing stage of our proposed method, we utilized the CLAHE algorithm to improve the contrast of blood vessels. In this section, we also investigate other new methods for improving image contrast. Vijayalakshmi et al. [61] proposed a method for improving the contrast of low-contrast images. The method involves two processes: detail information extraction and enhancing contrast through histogram equalization.

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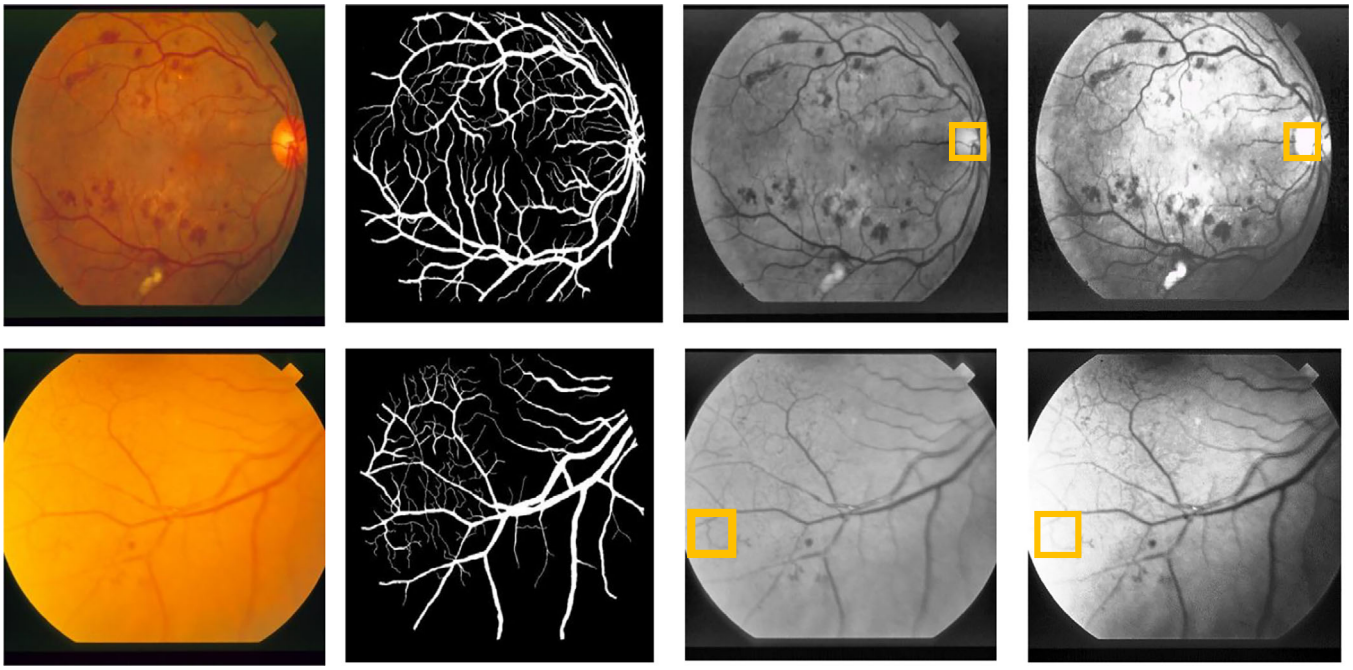


FIGURE 7 Comparison of CLAHE algorithm and the method proposed in [61] for enhancing the contrast of fundus images on DRIVE.

Initially, the detailed information is extracted through $l1 - l0$ minimization in multilevel decomposition. Subsequently, based on the background characteristics of the low-contrast image, the extracted detail layer is employed for enhancing contrast through histogram equalization. The paper utilizes optimization methods such as g alternating method of multiplier (ADMM) and specific tuning parameters to enhance image contrast through detail information and histogram equalization. Additionally, the proposed method suggests selecting appropriate parameters, such as regularization and brightness parameters, improving performance, and enhancing the final image quality. We implemented the methodology outlined in this paper. Subsequently, we compared its output with the CLAHE method on the DRVIE and STARE datasets. Some of the results obtained from these datasets are illustrated in Figures 7 and 8, respectively.

In another study, Vijayalakshmi et al. [62] suggested a method for enhancing image contrast based on Rènyi entropy. In this approach, the input image is initially partitioned into multiple segments, and then, by computing the two-dimensional histogram and utilizing Rènyi entropy, structural information about the image is extracted. Subsequently, using discrete functions derived from Rènyi entropy, a mapping function for enhancing image contrast is defined. In the subsequent steps, by imposing constraints on the discrete functions, density distributions and cumulative distribution functions (CDFs) are calculated. Finally, employing discrete cosine transformation (DCT) and weighting the DCT coefficients, local contrast enhancement is performed on the image. The method introduced in the paper utilizes the structural and colour information of the image, but it may encounter limitations when dealing with images containing intricate and densely packed vasculature and details. Additionally, if retinal images contain noise or other unintended deviations, this method may fail to provide satisfactory improvement and yield acceptable contrast enhancement results. On the other hand, CLAHE has shown better performance in enhancing vasculature contrast in retinal fundus images due to its better adaptation to the contrast diversity in images and noise control capabilities.

According to Figures 7 and 8, the method presented in the first paper may perform less than CLAHE in some situations, especially in areas of the fundus image that have very low contrast or a small contrast difference with the background. Also, this method often ignores the vessels that are present in the optic disc part of the retina. This could be attributed to two factors: Firstly, since the method under investigation utilizes the $l1 - l0$ minimization model for edge information extraction, it may not effectively preserve structural details in images with low contrast. Secondly, in regions with low contrast or minimal contrast difference with the background, the method may produce inaccurate results due to non-uniformity in pixel intensity distributions, potentially leading to the generation of artefacts or increased noise in the image. These are shown with yellow squares in these two figures. Instead, CLAHE, through a systematic histogram division process, prevents the creation of white and black spots in areas with different contrasts in the image. Moreover, CLAHE is capable of better adaptation to the contrast diversity in images and can perform better when

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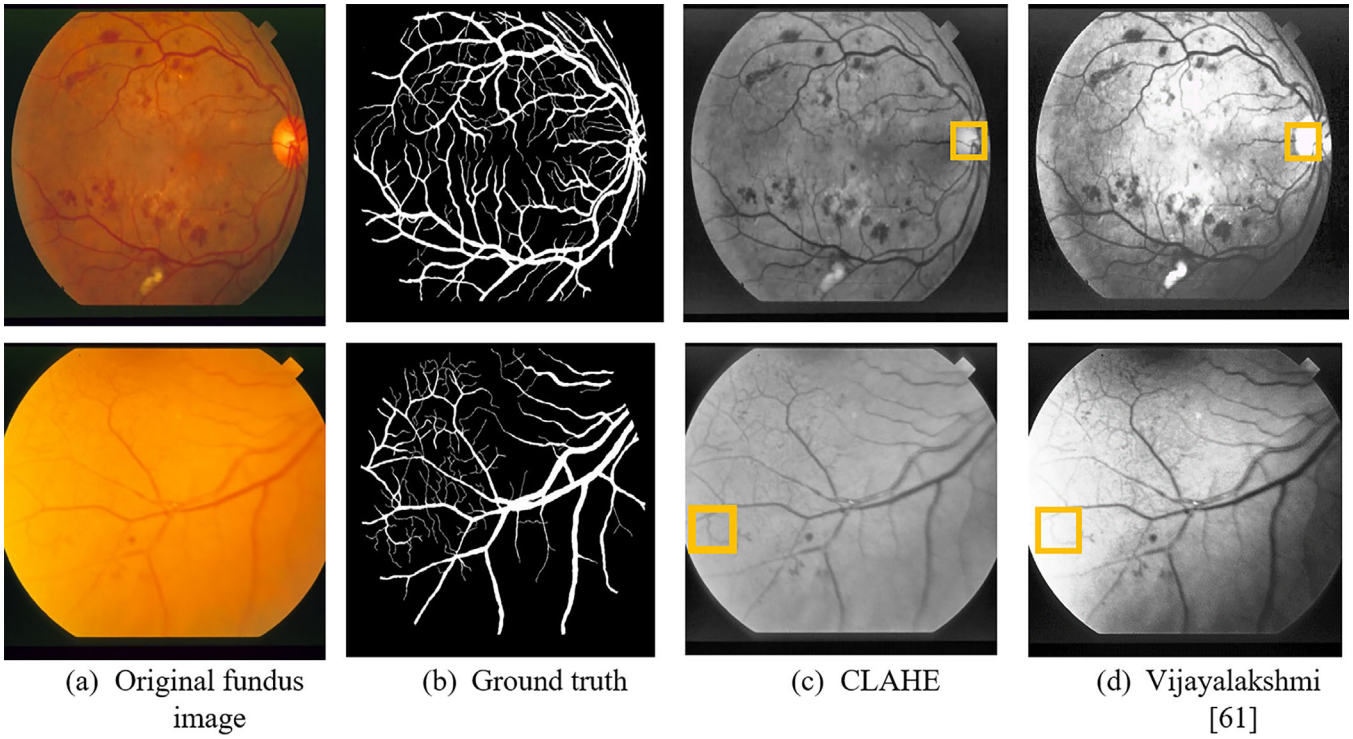


FIGURE 8 Comparison of CLAHE algorithm and the method proposed in [61] for enhancing the contrast of fundus images on STARE.

dealing with regions with low or different contrast from the background.

5.2 | Ablation study on model components

Following the hyper-parameter ablation study, we proceeded to assess the efficacy of the components integrated into our proposed approach.

5.2.1 | Pixel-level ablation study

To demonstrate the pixel-level impact, we disregarded the graph construction from images and conducted experiments from different perspectives. In one experiment, we trained the network without image patching and with image patching to understand the influence of patching in training. In another experiment, we examined the impact of the AB-FFM module in the network, conducting three sub-experiments to investigate individual components, including BConvLSTM, channel-wise attention, and pixel-wise attention.

Case study 1: Impact of the patching method

For our initial experiment, our objective is to showcase the influence of the patching technique on image segmentation accuracy and computation resources. Our approach involves training the model three times: first without patching the images, second with randomly patched input images. We then compare the segmentation outcomes on a random dataset. Although the patching strategies differ, all parameters remain constant. Figures 9 and 10 illustrate the qualitative segmentation outcomes for the DRIVE and STARE datasets for all three models.

The analysis of Figures 9 and 10 reveals that applying the patching technique before training is crucial to achieving effective segmentation of retinal vessels. Additionally, the patching approach outperforms without patching, particularly when fine and thick blood vessels intersect. Table 4 illustrates the quantitative evaluation outcomes.

In training the network with raw images, the network receives the entire image as an input and must use this information to learn the important features of the image. This can be difficult for the network, because crucial features of the image might be distributed across various regions of the image. In training with image patching, instead of receiving the whole image, the network receives a small patch of the image as input. The network should use these patches to learn the important features of the image. This can be easier for the network, as it only needs to use information from a small area of the image.

Table 4 shows that the patching strategy has improved all indices in the DRIVE and STARE datasets. This is because the network only needs to use the information of a small area of the image in training by patching images. This can make it easier for the network and help it learn the important features of the image more accurately.

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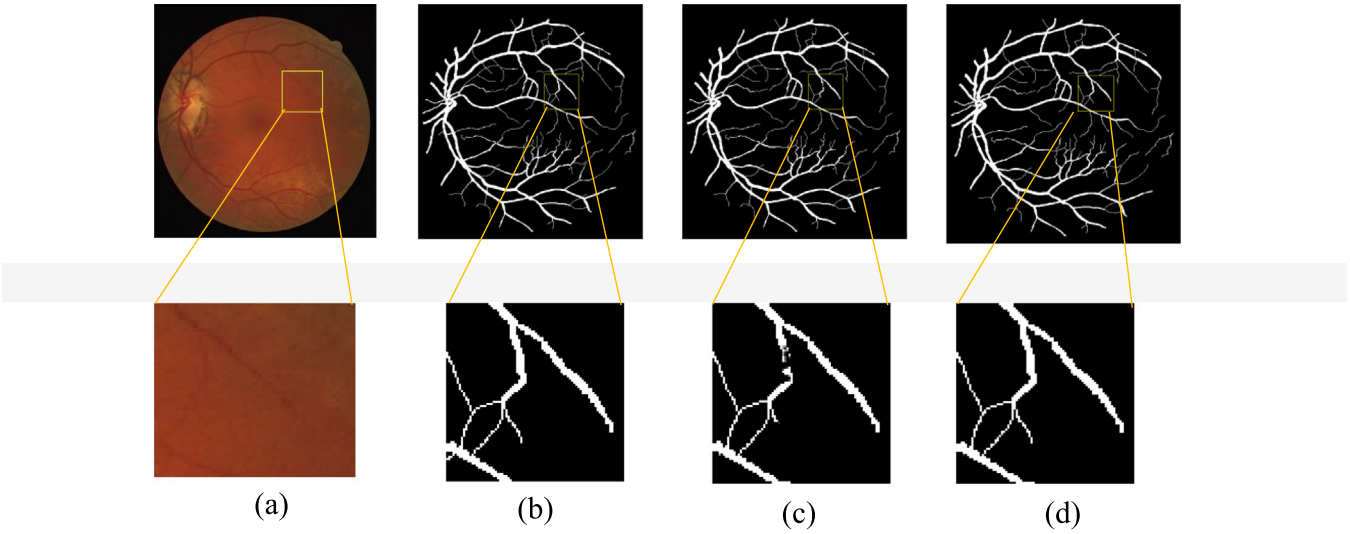


FIGURE 9 The DRIVE dataset's segmentation results using various patching strategies. (a) The initial fundus image, (b) the ground truth image, (c) the output of the segmentation without patching, and (d) the output of the segmentation with random patching.

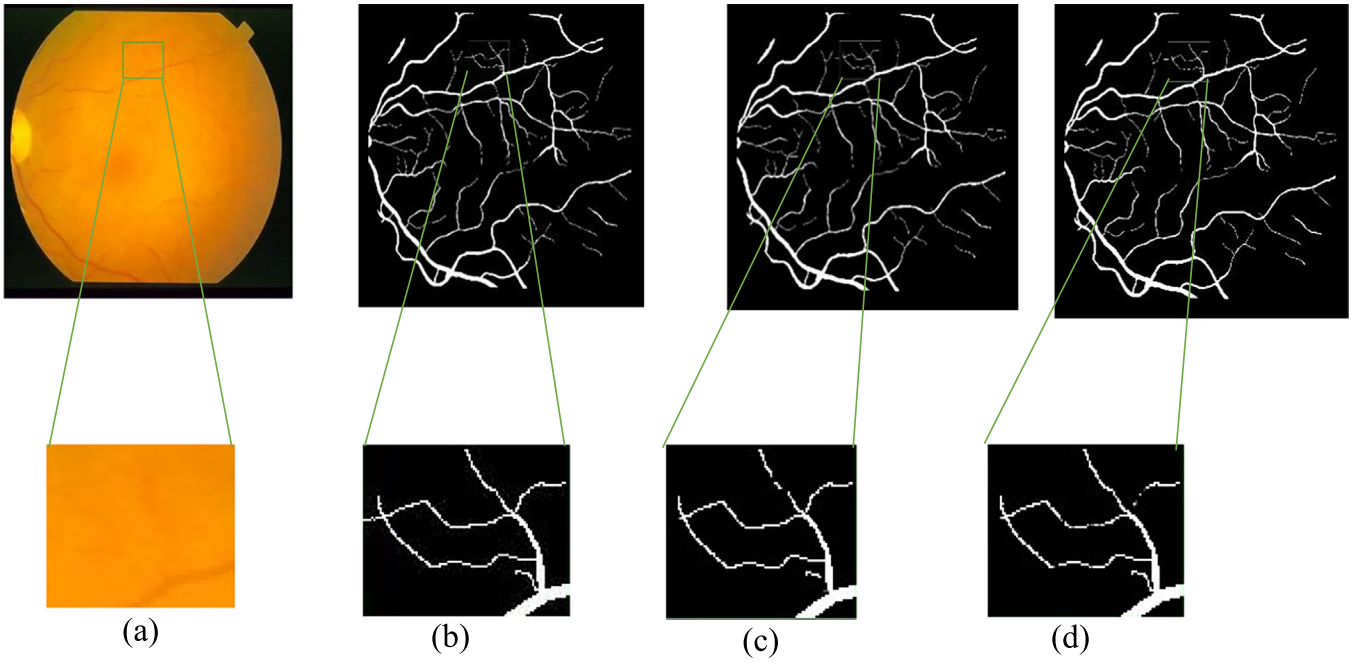


FIGURE 10 The STARE dataset's segmentation results using varied patching strategies. (a) The initial fundus image, (b) the ground truth image, (c) the output of the segmentation without patching, and (d) the outcome of the segmentation with random patching.

TABLE 4 The quantitative analysis comparing the segmentation results by different patching strategies on both the DRIVE and STARE datasets.

	DRIVE					STARE				
	Dice	**SP**	**SE**	**ACC**	**MCC**	**Dice**	**SP**	**SE**	**ACC**	**MCC**
Without patching	0.8261	0.9691	0.8532	0.9615	0.7823	0.8327	0.9611	0.8541	0.9719	0.7941
Random patching	0.8346	0.9723	0.8605	0.9714	0.7894	0.8459	0.9766	0.8701	0.9788	0.7986

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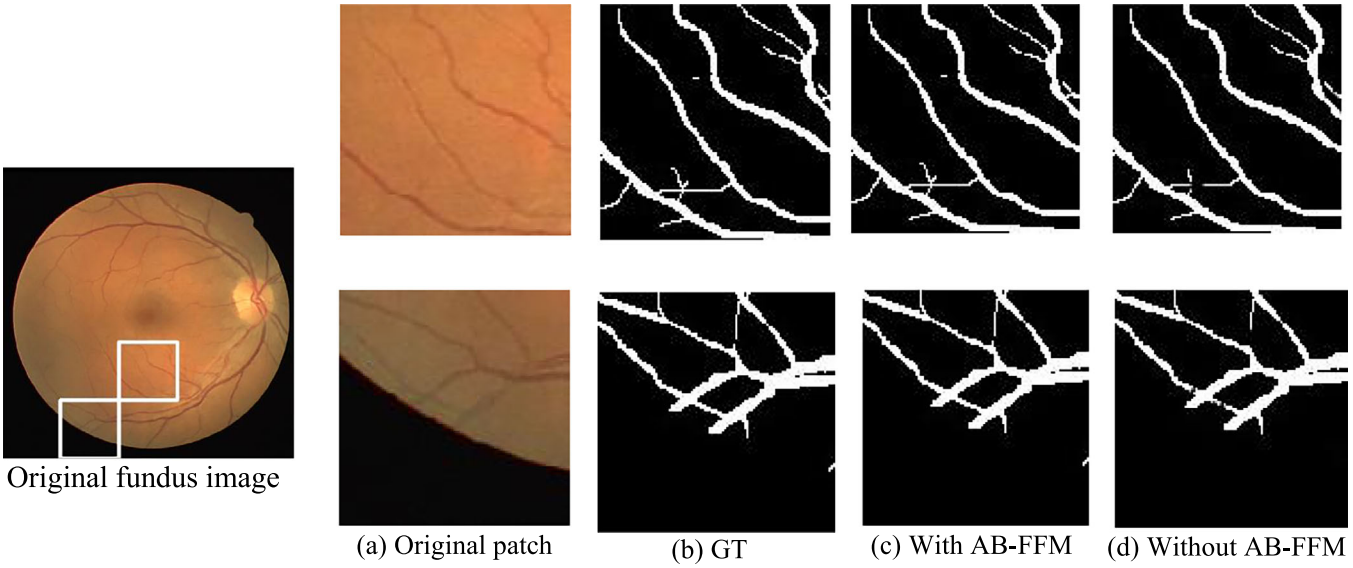


FIGURE 11 The impact of the attentional BConvLSTM feature fusion module (AB-FFM) on vascular segmentation in the DRIVE dataset.

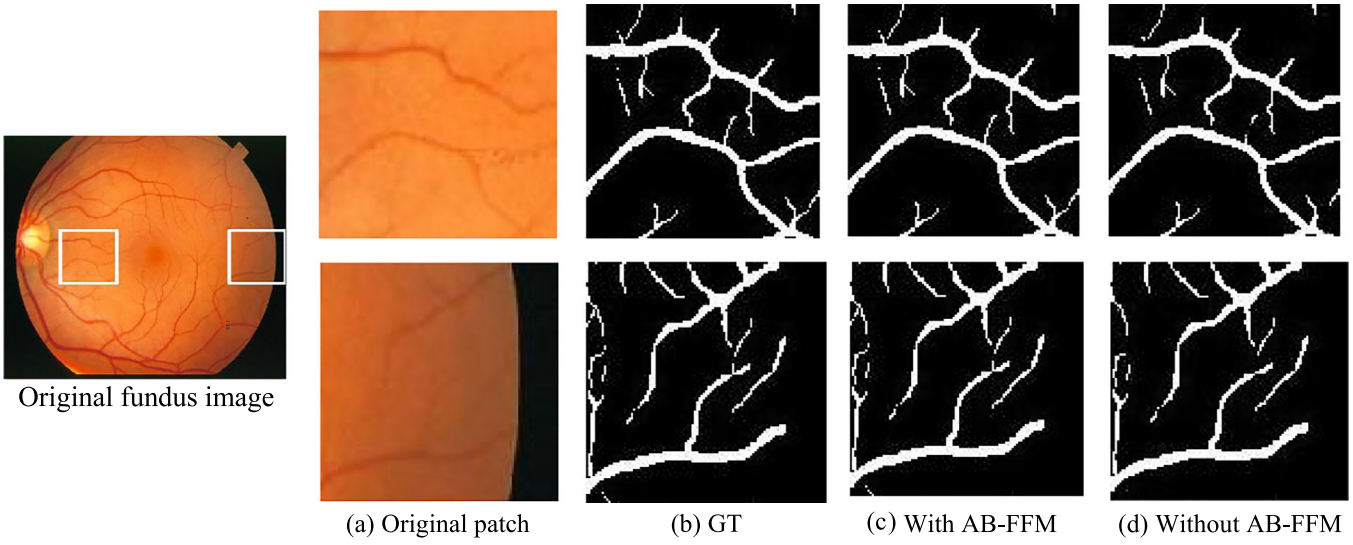


FIGURE 12 The impact of the attentional BConvLSTM feature fusion module (AB-FFM) on vascular segmentation in the STARE dataset.

Case study 2: Experiment of attentional BConvLSTM feature fusion module (AB-FFM)

In this section, we conducted an ablation study on the attentional BConvLSTM feature fusion module (AB-FFM) to assess the individual contributions and importance of different components of this module in enhancing the accuracy of feature fusion processes in the segmentation task. In Figures 11 and 12, we showcase the qualitative outcomes of segmenting images from the DRIVE and STARE datasets. Also, Table 5 illustrates the quantitative evaluation outcomes.

Full AB-FFM model. Initially, the full model incorporating all components of AB-FFM was employed as the baseline for comparison. This model utilizes the BConvLSTM module along with both pixel-wise and channel-wise attention mechanisms to integrate semantic information from various levels.

Pixel-wise attention removal. In this step, the pixel-wise attention component of AB-FFM was removed, leaving only the channel-wise attention mechanism. This step aimed to evaluate the importance of pixel-wise attention in improving accuracy and reducing noise in the model's output.

Channel-wise attention removal. In this stage, the channel-wise attention component of AB-FFM was excluded, retaining only the pixel-wise attention. This experiment was conducted to investigate the impact of channel-wise attention on recognizing the importance of semantic features in feature fusion.

BConvLSTM removal. In this phase, the BConvLSTM layers of AB-FFM were removed. This ablation was performed to assess the role of these layers in encoding semantic information from input features.

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TABLE 5 The quantitative analysis comparing the segmentation results by different attention modules on both the DRIVE and STARE datasets.

	DRIVE					STARE				
	Dice	**SP**	**SE**	**ACC**	**MCC**	**Dice**	**SP**	**SE**	**ACC**	**MCC**
Only PAM	0.8302	0.9733	0.8302	0.9729	0.7899	0.8399	0.9851	0.8705	0.9776	0.7982
Only CAM	0.8311	0.9745	0.8213	0.9736	0.7903	0.8402	0.9855	0.8712	0.9781	0.7987
CAM + PAM without BConvSLTM	0.8320	0.9759	0.8322	0.9738	0.7903	0.8410	0.9862	0.8718	0.9789	0.7988
CAM + PAM with BConvSLTM	0.8322	0.9764	0.8325	0.9743	0.7921	0.8412	0.9872	0.8720	0.9792	0.7991

Abbreviations: CAM, Channel attention module; PAM, Pixel attention module.

Based on Figures 11 and 12, the results indicate the significance of different segments of AB-FFM in enhancing the accuracy of feature fusion. The removal of pixel-wise attention led to increased noise in the output, while excluding channel-wise attention resulted in a reduction in the model's ability to discern semantic features. Additionally, removing BConvLSTM layers led to a loss of meaningful information encoded from input features.

The ablation study demonstrates that all three components pixel-wise attention, channel-wise attention, and BConvLSTM layers play crucial roles in improving the accuracy of retinal vessel segmentation. These findings provide a foundation for further optimization and the development of future model iterations.

5.2.2 | Graph-level ablation study

In this section, we integrated image-based graph construction and conducted several experiments. In one experiment, we focused on integrating the topological information of vessels with pixel-wise information in the segmentation process. In another experiment, we investigated the preservation of vessel structure and continuity in the segmentation process. Therefore, we investigate the impact of incorporating information from the GCN into the U-Net model for vascular segmentation. Two comparison scenarios will be examined: one involving the integration of information from GCN, and the other without the involvement of such information.

Case study 1: Incorporating information from GCN
The U-Net model is employed, integrating information obtained from the GCN module. This network performs segmentation by considering the topological and structural information of the vessels.

Case study 2: No incorporating Information from GCN
The U-Net model operates without integrating information from GCN. This case serves as a comparative analysis to evaluate the impact of graph-based information on vascular segmentation.

Figures 13 and 14 illustrate the qualitative impact of employing GCN on the segmentation results achieved on both the DRIVE and STARE. Additionally, Table 6 reports quantitative results on these two datasets.

The results of this study illustrate that the utilization of topological and structural information from the GCN significantly enhances precision, particularly sensitivity, in the segmentation of blood vessels in intricate areas and regions with intersections between fine and thick vascular structures. Moreover, the incorporation of the attention mechanism in GCN, along with skip connections, proves highly effective in preserving the continuity of pixel shapes within the vessels. Overall, these results affirm the noteworthy impact of incorporating graphical information on improving the accuracy and quality of the final vascular segmentation.

5.3 | Performance evaluation against existing approaches

Within this section, a qualitatively and quantitatively comprehensive evaluation was conducted to assess the proposed approach compared to other modern methodologies in this domain. Figures 15 and 16 present the qualitative segmentation outcomes of several advanced models, including our proposed method. These figures indicate that our model's segmentation outcomes exhibit the closest resemblance to the ground truth among the evaluated approaches. Also, the integrity of the segmentation is superior to that of other methods. Additionally, a quantitative comparison was performed by assessing our proposed method against eight other advanced methods, which consist of state of the art papers in this field. The quantitative comparison results for the DRIVE and STARE datasets can be found in Tables 7 and 8, respectively. The yellow boxes indicate regions of segmentation errors by methods.

A careful examination of Figure 15 reveals that, among graph-based methods, the scope method has not succeeded in accurately segmenting the central blood vessels. In contrast, our approach effectively addresses this challenge. Moreover, due to the utilization of pixel-wise attention mechanisms, our method demonstrates greater precision in separating individual pixels of fine vessels compared to the reference method. This observation is evident in both Figures 15 and 16.

Overall, it is clear that our proposed model surpasses other techniques in performance in accurately segmenting vessels, outstanding vessels or areas in the image near the background. Moreover, our method displays superior robustness and integrity compared to other proposed methods. Overall, the vessel segmentation outcomes of our method demonstrate

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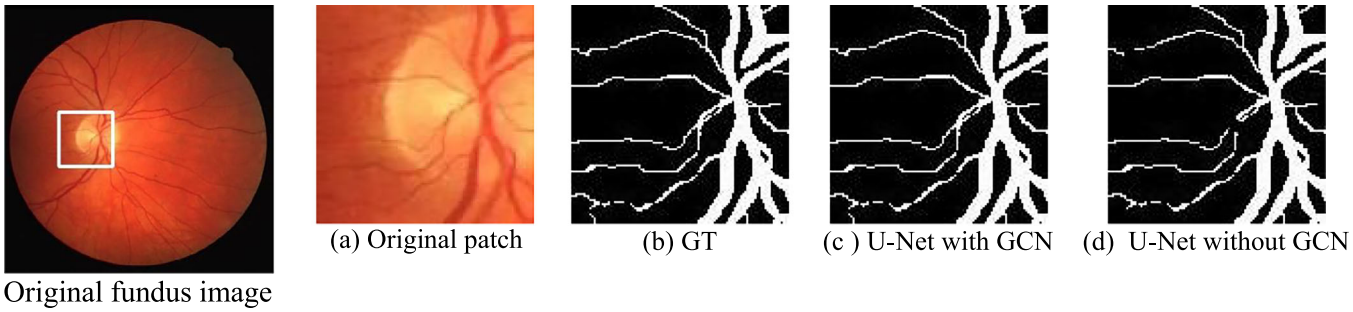


FIGURE 13 The impact of GCN on vascular segmentation in the DRIVE dataset.

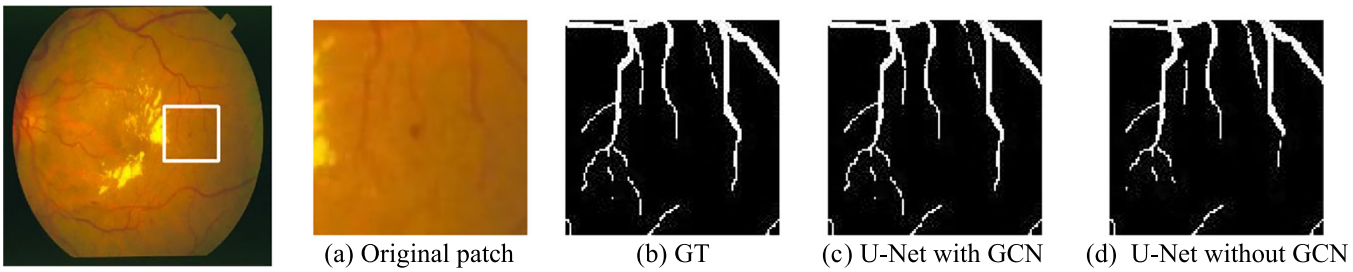


FIGURE 14 The impact of GCN on vascular segmentation in the STARE dataset.

TABLE 6 The quantitative analysis comparing the segmentation results by GCN on both the DRIVE and STARE datasets.

	DRIVE					STARE				
	Dice	**SP**	**SE**	**ACC**	**MCC**	**Dice**	**SP**	**SE**	**ACC**	**MCC**
U-Net with GCN	0.8351	0.9863	0.8422	0.9794	0.8035	0.8532	0.9883	0.8764	0.9819	0.8076
U-Net without GCN	0.8322	0.9764	0.8325	0.9743	0.7916	0.8412	0.9872	0.8720	0.9792	0.8021

Abbreviation: GCN, Graph convolution network.

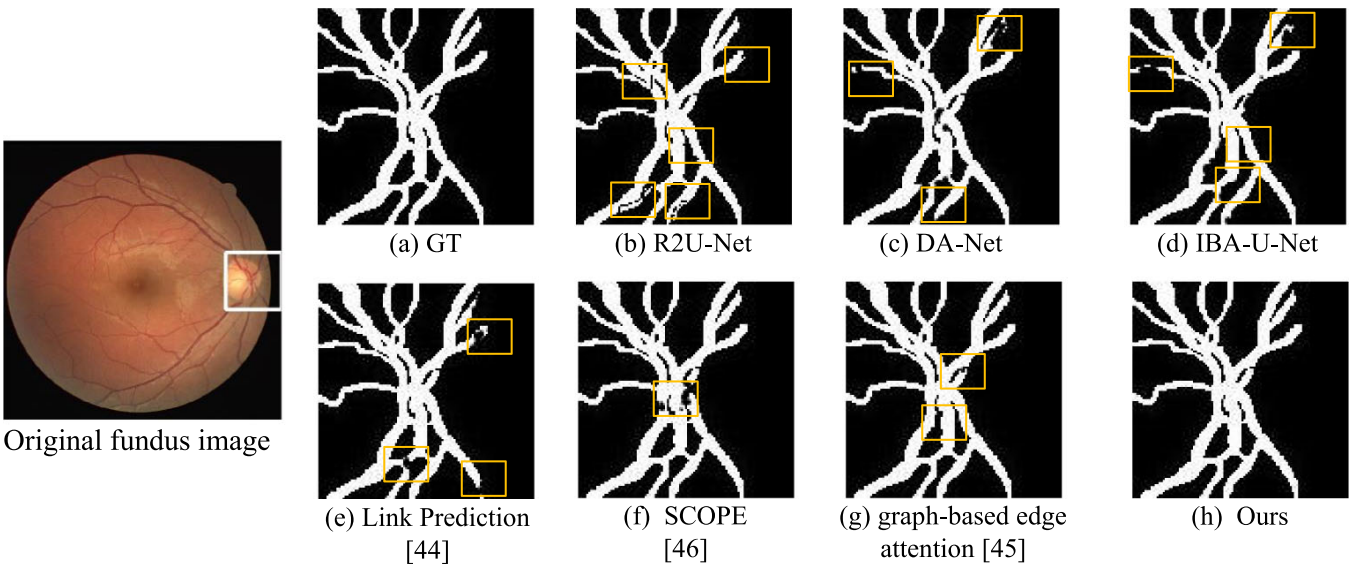


FIGURE 15 Segmentation performance comparison with our approach versus state-of-the-art methods on the DRIVE First row is about pixel-level approaches (b-d), and second row is about graph-level approaches (e-g).

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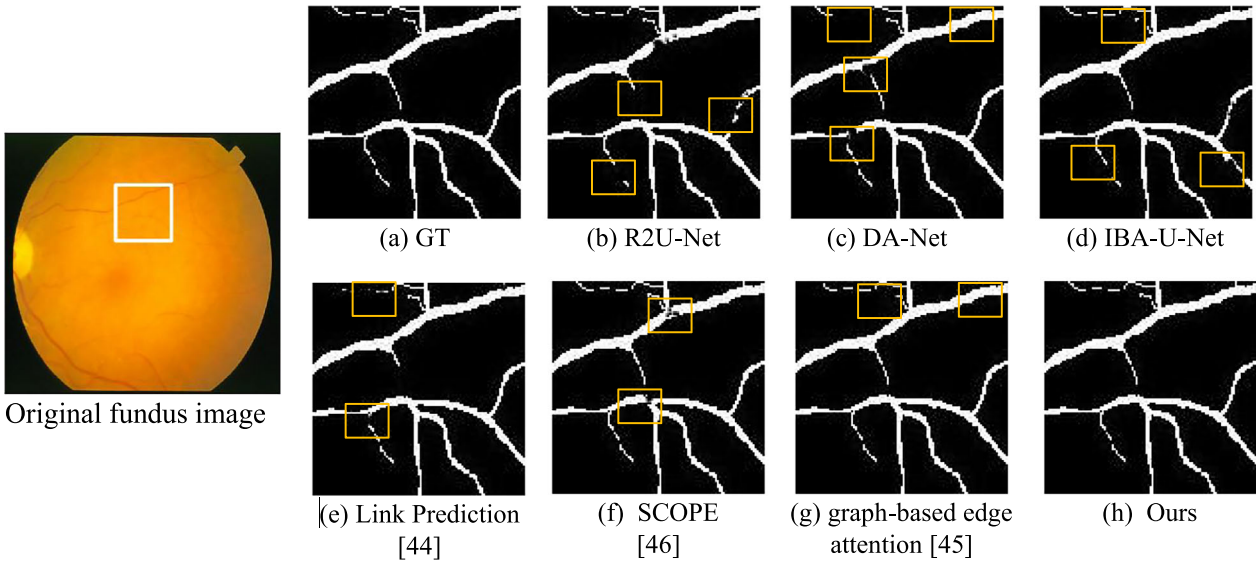


FIGURE 16 Segmentation performance comparison with our approach versus state-of-the-art methods on the STARE First row is about pixel-level approaches (b-d), and second row is about graph-level approaches (e-g).

TABLE 7 Comparative evaluation of state-of-the-art methodologies on the DRIVE dataset.

Model	Year	SP	SE	ACC	Dice	clDice	MCC
R2-UNet [30]	2018	0.9813	0.7792	0.9556	0.8171	–	–
U-Net [10]	2018	0.9820	0.7537	0.9531	0.8149	0.8147	0.7857
U-Net ++ [11]	2018	0.9861	0.8120	0.9710	0.8302	–	–
SAT-Net [63]	2021	**0.9870**	0.8117	0.9684	0.8174	–	–
IBA-U-Net [37]	2021	0.9703	0.7852	0.9550	0.8112	–	–
DA-Net [13]	2022	0.9818	0.7853	0.9565	0.8203	–	–
Link prediction [44]	2022	0.9812	0.8132	0.9677	0.8176	0.8267	–
IterMiUnet [27]	2023	0.9789	0.8053	0.9568	0.8262	–	–
NAUNet [64]	2023	0.9831	0.7999	0.9671	–	–	–
RADCU-Net [65]	2023	0.9807	0.8029	0.9582	–	–	–
PCRTAM-Net [66]	2023	–	0.8158	0.9710	0.830 5	–	–
SCOPE [46]	2023	0.9831	0.8377	0.9775	0.8278	0.8219	–
Graph-based edge attention [45]	2023	0.9850	0.84	0.9782	0.8345	0.8291	–
VGA-Net (ours)	2023	0.9863	**0.8422**	**0.9794**	**0.8351**	**0.8302**	**0.8035**

Abbreviations: DA-Net, Dual attention network; R2U-Net, Recurrent Residual Convolutional Neural Network; VGA-Net, Vessel graph based attentional U-Net for retinal vessel segmentation.

a significant level of similarity to the ground truth of the fundus images.

Tables 7 and 8 highlight the best performances in the evaluation criteria and display methods that utilized the patching approach in training in red. An analysis of the tables reveals that the VGA-Net method has attained the highest accuracy, sensitivity, Dice, and clDice scores across both the DRIVE and STARE. In the DRIVE dataset, our proposed method has exhibited enhancements in accuracy, sensitivity, Dice and clDice by 0.12%, 0.22%, and 0.06% and 0.11%, respectively, and 0.14%, 0.06%, 0.09 %, and 0.15% in STARE, compared to the sub-optimal approach. However, only the specificity parameter in the SAT-Net method is higher than in our method. This difference is attributed to the SAT-Net model. In SAT-Net, the original size image is directly employed as the input for the network, while we employed the patching technique in the proposed method.

6 | CONCLUSIONS

This paper introduces a network called VGA-Net, comprising both CNN and GCN components. Essentially, our approach integrates preprocessing techniques, graph convolution,

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TABLE 8 Comparative evaluation of state-of-the-art methodologies on the STARE dataset.

Model	Year	SP	SE	ACC	Dice	clDice	MCC
R2-UNet [30]	2018	0.9862	0.8298	0.9712	0.8475	–	–
U-Net [10]	2018	0.9842	0.8270	0.9690	0.8373	0.8324	0.7889
U-Net ++ [11]	2018	0.9843	0.8646	0.9753	0.8393	–	–
SAT-Net [63]	2021	**0.9927**	0.8072	0.9805	0.8270	–	–
IBA-U-Net [37]	2021	0.9832	0.7941	0.9750	0.8314	–	–
DA-Net [13]	2022	0.9864	0.8012	0.9668	0.8289	–	–
Link prediction [44]	2022	0.9814	0.8622	0.9766	0.8321	0.8326	–
IterMiUnet [27]	2023	0.9831	0.8069	0.9649	0.8231	–	–
NAUNet [64]	2023	0.9845	0.7951	0.9679	–	–	–
RADCU-Net [65]	2023	0.9831	0.8167	0.9682	–	–	–
PCRTAM-Net [66]	2023	–	0.8571	0.976 8	0.8464	–	–
SCOPE [46]	2023	0.9855	0.8693	0.9733	0.8479	0.8421	–
Graph-based edge attention [45]	2023	0.9879	0.8758	0.98	0.8523	0.8476	–
VGA-Net (ours)	2023	0.9883	**0.8764**	**0.9819**	**0.8532**	**0.8491**	**0.8076**

Abbreviations: DA-Net, Dual attention network; R2U-Net, Recurrent Residual Convolutional Neural Network; VGA-Net, Vessel graph based attentional U-Net for retinal vessel segmentation.

attention mechanisms, and advanced modules within the VGA-Net model to achieve highly effective performance against noise and various challenges. Through image preprocessing methods like background intensity normalization, CLAHE, and Un-sharp Masking filter, the quality of retinal images is significantly enhanced, extracting valuable information while eliminating noise and unnecessary details. Moreover, by employing CNN and GCN models, the method simultaneously extracts pixel-level features and captures relationships between neighbouring nodes in the graph, thereby enhancing detection accuracy and preserving both structural and pixel-wise vessel information. Additionally, advanced modules such as HDC and AB-FFM in the VGA-Net effectively reduce noise and discard irrelevant details, combining low-level and high-level semantic information to improve the accuracy of segmentation and withstand noise and artefacts. Overall, the proposed method showcases remarkable enhancements in retinal vessel detection accuracy while demonstrating robustness against noise and other artefacts.

Our proposed method leverages information at both pixel and graph levels, utilizing pixel-level details and the structural and topological information of retinal vessels for improved segmentation. It has demonstrated its effectiveness in successfully addressing the challenges related to preserving vessel shapes, segmenting intricate fine vessels, achieving complete segmentation in regions with a complex vascular network, and overcoming issues of over-segmentation and fractures in vessels along the central line of the retina. The evaluation of this method was performed on two datasets, specifically DRIVE and STARE, achieving the best outcomes in the segmenting process for fine and narrow vessels.

AUTHOR CONTRIBUTIONS

Yeganeh Jalali: Software development; writing—original draft; methodology design; validation; investigation; data curation. **Mansoor Fateh**: Supervision; conceptualization; visualization; writing—review and editing; software implementation. **Mohsen Rezvani**: Writing—review and editing; software development; formal analysis; resource allocation.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

We confirm the availability of the datasets utilized in this study, sourced from DRIVE and STARE repositories. The data can be accessed via the provided links. However, due to potential updates or changes in the repositories, we recommend referring to the following links for the most current and accurate access to the data.

DRIVE: [https://drive.grand-challenge.org]

STARE: [http://cecas.clemson.edu/~ahoover/stare/probing/index.html].

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